Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/13/2017
PI Name:	Hyers, Robert Ph.D.		
Project Title:	Modeling and Simulation of Electrostatically Levitated Multiphase Liquid Drops		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01609-2280	**Congressional District:**	2
Comments:	Formerly with University of Massachusetts at Amherst.		
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Japanese Space Agency (JAXA) AO for International Research Participation on ISS Kibo Module
Start Date:	04/08/2014	**End Date:**	04/07/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Craven, Paul	**Contact Phone:**	256-544-7649
Contact Email:	pcravenRRR@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lee, Jonghyun Ph.D. (University of Massachusetts) Watanabe, Masahito Ph.D. (JAXA PI: Gakushuin University)		
Grant/Contract No.:	NNX14AH85G		
Performance Goal No.:			
Performance Goal Text:			

Funding is for Dr. Hyers's role as U.S. Principal Investigator (PI) on Professor Masahito Watanabe's (Department of Physics, Gakushuin University, Tokyo, Japan) proposal, “Interfacial Phenomena and Thermal Properties of High-Temperature Melts,” funded by the Japanese Aerospace Exploration Agency (JAXA). Prof. Masahito Watanabe's project: During steeling making process, such as continuous casting, the impurity in the cast steel is influenced by the interplay between the molten steel and molten slags. Understanding the interfacial phenomena is necessary to produce high-purity steels. An accurate measurement of interfacial tension between molten steel and molten slag is one essential step to understand the interfacial phenomena. Well-controlled oscillation of a sample with a concentric core (molten steel) – shell (molten slag) structure is required to measure the interfacial tension. Due to the effect of gravity, this concentric arrangement cannot be obtained using ground-based facilities. An Electrostatic Levitation Furnace (ELF) in

Task Description:	the KIBO deck in the International Space Station (ISS) will be utilized for the measurement of interfacial tension at the interface between molten steel and molten slag. For the success of the space mission, it is of critical importance to predict the interfacial tension and the influence of main parameters, such as the core-shell ratio, thermophysical properties of molten slags, magnitude of initial deformation, magnitude of electrostatic forces, and off or inphase oscillation, on the predicted interfacial tension. With such information, the space experiment can be designed efficiently and effectively. Dr. Hyers's (University of Massachusetts) scope of work as part of Dr. Watanabe's proposal is summarized below: 1. Simulation of the flow in electrostatically levitated core-shell droplets, including dynamics such as oscillations and viscous effects on the dynamics. 2. Simulation of the effect of changes in the dimensions of the core and shell on the surface oscillations. 3. Simulation of the time dependence of the changes in surface oscillations due to reactions between the oxide and steel phases. 4. Design, direct, and analyze experiments to be conducted in the Marshall Space Flight Center (MSFC) Electrostatic Levitation Facility. The Research is to use data from an Electrostatic Levitator (ESL) facility supplied by the Japanese to be used on the International Space Station, and also a ground based ESL facility located at MSFC.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of the project are expected to be utilized for more efficient production of high-purity steels, which are the main metals used on Earth.
Task Progress:	Highlights of Work Accomplished During Grant Period 2016-17. • Face-to-face meeting with JAXA PI Prof. Watanabe to discuss latest experimental results from parabolic flight campaign. A novel phenomenon was observed, where a metal-slag droplet did not form the usual core-shell structure. Instead, the droplet formed a double droplet. From the images, Prof. Watanabe's group was able to determine the interfacial energy between the metal and slag phases. UMass is modeling this case and its dynamics. • A single phase droplet oscillation was successfully modeled with the volume of fluid method using ANSYS Fluent and it is being extended to simulate multiphase droplets. • Attended the 145th Minerals, Metals & Materials Society (TMS) Annual Meeting in Nashville, TN, February, 2016. Status of ELF • The main part of ELF was launched on 8/19/2015 by HTV-5 (JST, Tanegashima, Japan). • The remaining part of ELF was launched by Cygnus (OA-4, USA). • ELF assembly in the International Space Station started on 1/12/2016. • Measured interfacial tension between Fe and welding flux / Fe and BF slag in ELF on Parabolic Flights in May 2016. • Melted two samples (Zr under Ar Atmosphere, Er-doped Calcium Aluminate under N2 Atmosphere) in December 2016. • Plan to check the capability of droplet oscillation and process oxide samples before April 2017.
Bibliography Type:	Description: (Last Updated: 03/30/2026)
Abstracts for Journals and Proceedings	Lee J, Sumaria K, Baldwin E, Mooney K, Watanabe M, Hyers RW. "Numerical Simulation of the Oscillation and Damping of Core-Shell Structured Iron-Slag Droplets for the Measurements of Surface Tension and Viscosity in Reduced Gravity." TMS 2016. 145th Annual Meeting, The Minerals, Metals and Materials Society, Nashville, TN, February 14-18, 2016. The 145th TMS Annual Meeting, Nashville, TN, Feb. 2016. TMS 2016. 145th Annual Meeting, The Minerals, Metals and Materials Society, Nashville, TN, February 14-18, 2016. http://www.programmaster.org/PM/PM.nsf/ApprovedAbstracts/B03068DAEA913BE285257E74004D0700?OpenDocumer ; accessed 2/15/2017. , Feb-2016
Books/Book Chapters	Allanore A, Bartlett L, Wang C, Zang L, Lee J. (Eds.) "EPD Congress 2016. (Ed. note: Annual collection of conference proceedings that addresses extraction and processing metallurgy. The papers in this book are drawn from symposia held at the 2016 Annual Meeting of The Minerals, Metals & Materials Society.)" Ed. A. Allanore, L. Bartlett, C. Wang, L. Zang, J. Lee. Hoboken, New Jersey : Wiley & Sons Inc., 2016., Feb-2016